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Marzo, Roy Rillera; Li, Man ; Yang, Qizhi ; Chen, Hana; Sam, Rany ; Farooqui, Maryam

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Projected Burden of Alzheimer's Disease and Other Dementias in the Western Pacific, 2023–2050

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Abstract:	Abstract Background: Alzheimer's disease and other dementias (ADODs) are rising rapidly as populations age, but region-specific forecasts for the Western Pacific are limited. We projected disability-adjusted life-years (DALYs) and deaths to 2050 and evaluated alternative futures that modify environmental, behavioural/metabolic, and early-life risks. Methods: Using the Global Burden of Disease (GBD) 2021 scenario-based framework, we modelled ADOD burden for 37 Western Pacific countries/areas from 2023–2050, stratified by age and sex. Primary outcomes were all-age rates per 100,000 for DALYs and deaths; secondary outputs included age-distribution of burden and national rankings. Projections incorporated reference and four counterfactual scenarios. Uncertainty was quantified with 1,000 Monte Carlo draws and summarised as means with 95% uncertainty intervals (UIs). Findings: Regional all-age DALY rates rise from 777.6 (95% UI 375.5–1,714.8) in 2023 to 1,980.9 (964.7–4,176.9) in 2050 (+154.7%), while death rates increase from 41.1 (10.5–110.2) to 119.7 (30.5–302.8) (+191.3%). Female rates exceed male rates throughout, widening the absolute gap. The burden concentrates in very old ages: by 2050, those

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	aged 80–94 account for ~62% of DALYs and ~69% of deaths. Japan remains highest in level, Korea converges, and China and Singapore show the steepest absolute increases. Scenario curves overlap until the 2040s; by 2050, the Combined package yields slightly higher rates than the Reference due to survival-driven composition effects. Conclusion: Demographic ageing will dominate dementia burden in the Western Pacific. Preventive efforts may delay onset but cannot offset major growth in service needs, underscoring the urgency of dementia-capable, gender-responsive health and social care expansion.



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**Projected Burden of Alzheimer's Disease and Other Dementias in the Western Pacific,
2023–2050**

*Roy Rillera Marzo

Faculty of Humanities and Health Sciences

Curtin University, Malaysia, Miri, Sarawak, Malaysia

roy.marzo@curtin.edu.my

*Man Li, Department of Intensive Care Unit, The First Affiliated Hospital of Jinan University, Guangzhou, China. 931329435@qq.com

Qizhi Yang

Northern Jiangsu People's Hospital Affiliated to Yangzhou University, Yangzhou, Jiangsu, China, 225009

E-mail: qizhiyang0504@163.com

Hana Chen Wei Jun

Faculty of Humanities and Health Sciences

Curtin University, Malaysia, Miri, Sarawak, Malaysia

hanachen@curtin.edu.my

Rany Sam

National University of Battambang, Battambang Province, Cambodia

sam.rany@nubb.edu.kh

Maryam Farooqui

School of Pharmaceutical Sciences, Universiti Sains Malaysia, Pulau Pinang, Malaysia

farooquimaryam77@gmail.com

Manah Chandra Changmai

Centre for Anatomy and Human Identification, University of Dundee, Scotland, UK

mchangmai001@dundee.ac.uk

Zhumabekova Bibigul

Doctor of Biological Sciences, Margulan University, Pavlodar, Kazakhstan.

zhumabekovab@ppu.edu.kz

Aubakirova Karlygash
Department of Biotechnology and Microbiology, L.N. Gumilyov Eurasian National
University, Astana, Kazakhstan;
aubakirova_km@enu.kz

Titik Respati
Department of Public Health, Faculty of Medicine Universitas Islam Bandung, Indonesia
titik.respati@gmail.com

Dandan Xu
Department of Emergency intensive care unit, The Affiliated Hospital of Xuzhou Medical
University, Xuzhou, China
E-mail: 214731522@qq.com

Hongpeng Zhang
The First Affiliated Hospital of Jinan University, 613 Huangpu Avenue West, Guangzhou
510630, China
zhp1981@jnu.edu.cn

Zelalem Meraf Wolde
Department of Statistics, Injibara University, Injibara, Ethiopia
zdmeraf23@gmail.com

Wegayehu Sheferaw, Mizan Tepi University, Ethiopia and University of Canberra, Canberra,
Australia wegayehu.info@gmail.com

Corresponding author

Zelalem Meraf Wolde and Hongpeng Zhang

*Roy Rillera Marzo and Man Li have made equal contributions to this work

Abstract

Background: Alzheimer’s disease and other dementias (ADODs) are rising rapidly as
populations age, but region-specific forecasts for the Western Pacific are limited. We

projected disability-adjusted life-years (DALYs) and deaths to 2050 and evaluated alternative futures that modify environmental, behavioural/metabolic, and early-life risks.

Methods: Using the Global Burden of Disease (GBD) 2021 scenario-based framework, we modelled ADOD burden for 37 Western Pacific countries/areas from 2023–2050, stratified by age and sex. Primary outcomes were all-age rates per 100,000 for DALYs and deaths; secondary outputs included age-distribution of burden and national rankings. Projections incorporated reference and four counterfactual scenarios. Uncertainty was quantified with 1,000 Monte Carlo draws and summarised as means with 95% uncertainty intervals (UIs).

Findings: Regional all-age DALY rates rise from 777.6 (95% UI 375.5–1,714.8) in 2023 to 1,980.9 (964.7–4,176.9) in 2050 (+154.7%), while death rates increase from 41.1 (10.5–110.2) to 119.7 (30.5–302.8) (+191.3%). Female rates exceed male rates throughout, widening the absolute gap. The burden concentrates in very old ages: by 2050, those aged 80–94 account for ~62% of DALYs and ~69% of deaths. Japan remains highest in level, Korea converges, and China and Singapore show the steepest absolute increases. Scenario curves overlap until the 2040s; by 2050, the Combined package yields slightly higher rates than the Reference due to survival-driven composition effects.

Conclusion: Demographic ageing will dominate dementia burden in the Western Pacific. Preventive efforts may delay onset but cannot offset major growth in service needs, underscoring the urgency of dementia-capable, gender-responsive health and social care expansion.

Keywords: Western Pacific; Alzheimer's disease; dementia; projections; disability-adjusted life-years; mortality; ageing; scenarios; health systems; caregiver support

Introduction

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Dementia—principally Alzheimer’s disease and other dementias (ADODs)—has become a defining public-health challenge of population ageing. Globally, more than 55 million people are living with dementia today, and the number is projected to reach ~139 million by 2050, with the fastest growth occurring in low- and middle-income settings (Alzheimer’s Disease International [ADI], 2024). Independent GBD forecasts likewise project substantial increases to 2050 (GBD 2019 Dementia Forecasting Collaborators, 2022). The economic toll is already immense—exceeding US\$1.3 trillion in 2020 and expected to approach US\$2.8 trillion by 2030 (ADI, 2024). Although improved survival from cardiovascular and infectious conditions has delivered gains in longevity, it simultaneously enlarges the pool of adults who survive into ages at which dementia incidence and case fatality are highest (Li et al., 2022).

Nowhere are these dynamics more consequential than in the Western Pacific Region (WPR). The WPR is undergoing rapid demographic transition: >245 million adults aged ≥65 years already live in the region, and this number is expected to double by 2050 (Gong et al., 2024). The region’s epidemiologic profile further amplifies risk: many WPR countries have above-global-average exposure to modifiable determinants—diabetes, obesity, harmful alcohol use, physical inactivity, and tobacco—alongside persistent hypertension and elevated fasting glucose (Gong et al., 2024; Li et al., 2022). WHO guidance underscores that acting on these risks across the life course (e.g., blood-pressure control, smoking cessation, physical activity, hearing care, and education that builds cognitive reserve) can delay onset and compress disability, but will not fully offset the arithmetic of ageing (World Health Organization [WHO], 2019, 2023).

Health-system preparedness remains uneven. Only a minority of WPR jurisdictions have comprehensive dementia strategies in place: six countries maintain standalone national dementia plans and five embed dementia within broader policies, leaving large parts of the region without structured approaches to prevention, diagnosis, long-term care, and caregiver support (Timmins et al., 2024). This readiness gap is particularly concerning given marked between-country heterogeneity in projected growth—ranging from comparatively slower increases in super-aged settings to steep escalations in rapidly ageing, populous countries (Gong et al., 2024; Li et al., 2022).

Against this backdrop, robust, region-specific projections are essential for planning. Using the Global Burden of Disease (GBD) 2021 scenario-based forecasting framework, building on the GBD 2019 dementia forecasting approach (GBD 2019 Dementia Forecasting

Collaborators, 2022), this study estimates all-age and age-specific DALY and death rates for ADODs in the WPR from 2023 to 2050, disaggregated by sex and age; maps national and subregional patterns; and evaluates alternative futures that target environmental, behavioural/metabolic, and early-life determinants. Our objective is to provide decision-makers with policy-ready evidence on when, where, and for whom dementia burden will rise, so that prevention, service capacity, financing, and gender-responsive caregiver supports can be scaled proactively across the Western Pacific.

Methods

Overview

We analysed trends and projections of deaths and disability-adjusted life-years (DALYs) attributable to Alzheimer's disease and other dementias in the Western Pacific region using the Global Burden of Disease 2021 (GBD 2021) forecasting ("Foresight") framework developed by the Institute for Health Metrics and Evaluation (IHME). The study follows the GATHER reporting guidelines. All inputs are aggregated GBD data without personally identifiable information; therefore, ethics review and individual consent were not required. Data and model outputs (subject to IHME data use terms) and analysis code are publicly available; data were accessed on 25 August 2025.

Data sources, case definition, and geography

Cause estimates correspond to the GBD cause "Alzheimer's disease and other dementias", mapped from ICD-10 codes G30, F01, F02, F03. We extracted regional and national estimates for the Western Pacific set used in our analysis; the full country/area list appears in **Table 1**. To avoid ambiguity between WHO/WPRO membership and GBD regional groupings, we provide both the country list and the data source identifiers used for extraction.

Table 1. Country/area classification in the Western Pacific Region

Category	Country/Area
High-Income and Core East Asian Countries/Areas	Japan; Republic of Korea; China; Singapore; Australia; New Zealand; Taiwan, China; Brunei Darussalam
Southeast Asian Countries	Viet Nam; Malaysia; Philippines;

	Cambodia; Lao People's Democratic Republic; Micronesia (Federated States of); Mongolia
Pacific Island Countries (Melanesia, Micronesia, Polynesia)	Palau; Cook Islands; Niue; Tonga; Tuvalu; Fiji; Kiribati; Marshall Islands; Nauru; Papua New Guinea; Samoa; Solomon Islands; Tokelau; Vanuatu

Measures and stratification

Primary outcomes were DALYs per 100,000 population and deaths per 100,000 population, reported as all-age rates and as counts. We stratified results by sex (female, male, both) and age (5-year groups from 0–4 to 95+). Where noted, we also present age-standardised rates using the GBD standard population; age-standardisation choices are stated in the relevant figure captions. Uncertainty is expressed as 95% uncertainty intervals (UIs) derived from 1,000 posterior draws.

Scenario-based forecasting framework

We adopted IHME’s GBD 2021 scenario framework to project annual outcomes for 2023–2050. The framework integrates historical burden, time-varying covariates (e.g., Socio-demographic Index [SDI], Summary Exposure Values [SEVs] for relevant risks, educational attainment, and metabolic risk measures), and scenario-specific trajectories. We report five scenarios used in our figures: Reference, Safer Environment, Improved Behavioral and Metabolic Risks, Improved Childhood Nutrition and Vaccination, and Combined. Projections are generated by an ensemble of statistical models (including autoregressive and Bayesian regression components) selected and weighted by predictive performance. The generic model can be written as

$$\log Y_{c,a,s}(t) = \beta_0 + \sum_k \beta_k X_{c,a,s,k}(t) + \gamma_S + \epsilon,$$

where $Y_{c,a,s}(t)$ is the rate for country c , age group a , sex s , and year t ; $X_{c,a,s,k}(t)$ are covariates with scenario-specific trajectories; and γ_S indexes scenario effects. Future covariates follow IHME’s reference assumptions with scenario-appropriate adjustments. We summarise posterior means as point estimates with 95% UIs from the draws.

Statistical analysis and visualisation

We summarised regional time-series for DALY and death rates (Figure 1), age-specific counts and rates (Figure 2), national rate levels and rank changes (Figure 3), spatial distributions in 2023 and 2050 using identical class breaks and colour scales (Figure 4), and scenario-specific trajectories (Figure 5). For rank-change plots, ranks are computed from posterior mean rate. All analyses were performed in R 4.2.1.

Uncertainty, sensitivity, and validation

Primary uncertainty reflects posterior variance from the IHME forecasting draws. We propagated this uncertainty when calculating regional aggregates and scenario differences. As a sensitivity analysis, we repeated key summaries using age-standardised rather than all-age rates to distinguish compositional ageing from changes in age-specific risk. We also compared 2015–2019 hindcasts against held-out observations to assess calibration; and performance metrics.

Ethics approval and consent to participate: Not applicable. This study used aggregated, de-identified data from publicly available sources (Global Burden of Disease) and involved no interaction with human participants or access to identifiable personal data.

RESULTS

Figure 1 displays all-age rates per 100,000 for the Western Pacific and shows a steep, sustained rise from 2023 to 2050. For both sexes combined, the DALY rate increases from 777.6 (95% UI 375.5–1,714.8) in 2023 to 1,980.9 (964.7–4,176.9) in 2050—an absolute gain of 1,203.3 (+154.7%; CAGR $\approx$ 3.52%/year). Over the same period the death rate rises from 41.1 (10.5–110.2) to 119.7 (30.5–302.8)—an absolute gain of 78.6 (+191.3%; CAGR $\approx$ 4.04%/year), indicating that mortality accelerates faster than disability. Sex-specific trajectories are parallel but shifted upward for women: female DALYs increase from 1,050.2 to 2,713.2 (+158.3%), male DALYs from 513.5 to 1,251.8 (+143.8%); the female–male DALY gap widens from 536.7 to 1,461.4 (+924.7). Female deaths grow from 58.0 to 169.2 (+191.7%), male deaths from 24.8 to 70.5 (+184.9%); the female–male death gap widens from 33.2 to 98.7 (+65.5). Uncertainty intervals widen toward 2050, but the direction and magnitude of growth are robust across the forecast horizon. Taken together, the faster rise in

deaths than DALYs and the expanding female predominance signal increasing late-life care needs and the imperative to prioritize service capacity and social support for older women in super-aged Western Pacific health systems.

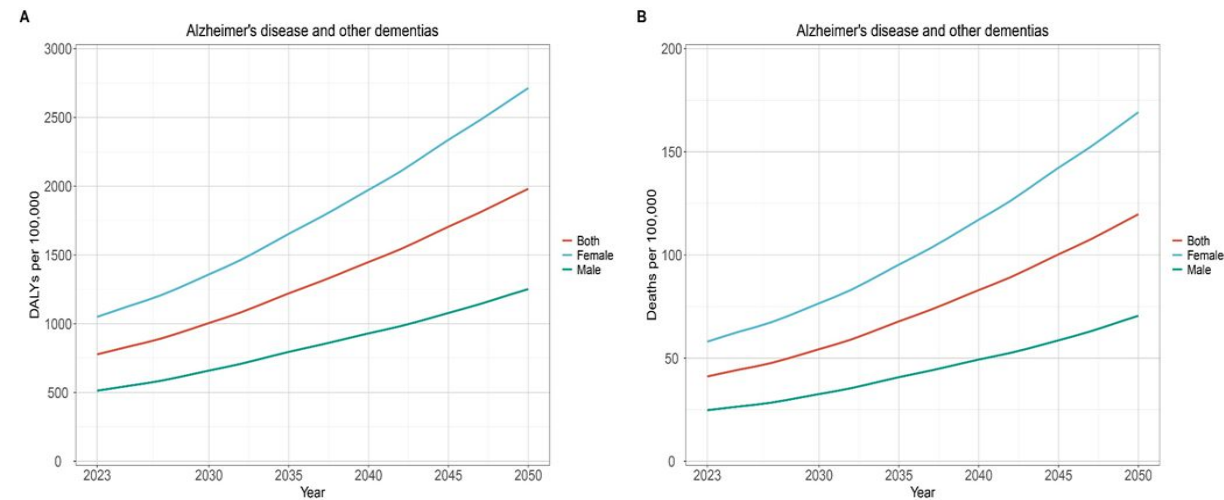


Figure 1. Estimated trends in the rates of DALYs (A) and deaths (B) due to Alzheimer's disease and other dementias in the Western Pacific region, 2023–2050. DALYs = disability-adjusted life-year

Figure 2 disaggregates the Western Pacific dementia burden by age and sex, with panels A/B/D/E showing both-sex total counts and panels C/F combining sex-specific counts (bars) with age-specific rates per 100,000 (lines); shaded bands and error bars denote 95% uncertainty intervals.

(A) Absolute total DALYs (counts) rise steeply with age and are overwhelmingly concentrated in the oldest-old. The largest increases are at 85–89 (~5.83 million) and 80–84 (~5.81 million), followed by 90–94 (~3.21 million) and 95+ (~2.91 million). Ages 40–59 show net declines, indicating that the regional expansion of disability is driven almost entirely by cohorts entering ≥ 70 years, with 75–79 also contributing materially. Error bars indicate 95% UIs and widen at advanced ages.

(B) Percentage growth mirrors panel A but emphasises the intensity at extreme ages: DALYs increase by ~364% at 95+, ~216% at 85–89, ~192% at 80–84, ~185% at 90–94, and ~105% at 75–79. Most bands below 60 decline. Thus, both the rate of increase and the absolute uplift peak in very old age.

(C) By 2050, the age composition of DALYs is dominated by the oldest-old: 80–94 account for ~62.2% and 95+ for ~10.3% (combined ~72.5%), whereas ≤ 69 contribute ~8.4%. In 2023, the share at ≥ 80 was $\approx 60\%$, evidencing a marked upward shift. The overlaid age-specific DALY rates per 100,000 increase monotonically with age and are higher in women; female-to-male rate ratios rise across ages, exceeding 2 by 85–89. Shaded ribbons are 95% UIs.

(D) Deaths display the same ageward shift. The largest absolute increases from 2023 to 2050 occur at 85–89 (~399,547), 80–84 (~308,162), 95+ (~288,277), and 90–94 (~271,104), while 40–59 decline slightly. Hence, nearly all future growth in dementia mortality accrues at ≥ 80 years (with widening UIs at the oldest ages).

(E) Percentage increases in deaths are most pronounced at the extremes: ~356% at 95+, ~208% at 85–89, ~186% at 80–84, ~179% at 90–94, and ~100% at 75–79. Declines below 60 again reflect demographic composition rather than direct evidence of falling age-specific risk.

(F) In 2050, deaths are highly concentrated at advanced ages: 80–94 contribute ~68.6% and 95+ ~17.0%, whereas ≤ 69 account for ~3.1%. Age-specific death rates per 100,000 escalate sharply with age and are higher in women; female-to-male rate ratios rise with age and surpass 2 by 85–89. Ribbons denote 95% UIs, which broaden at the extreme ages.

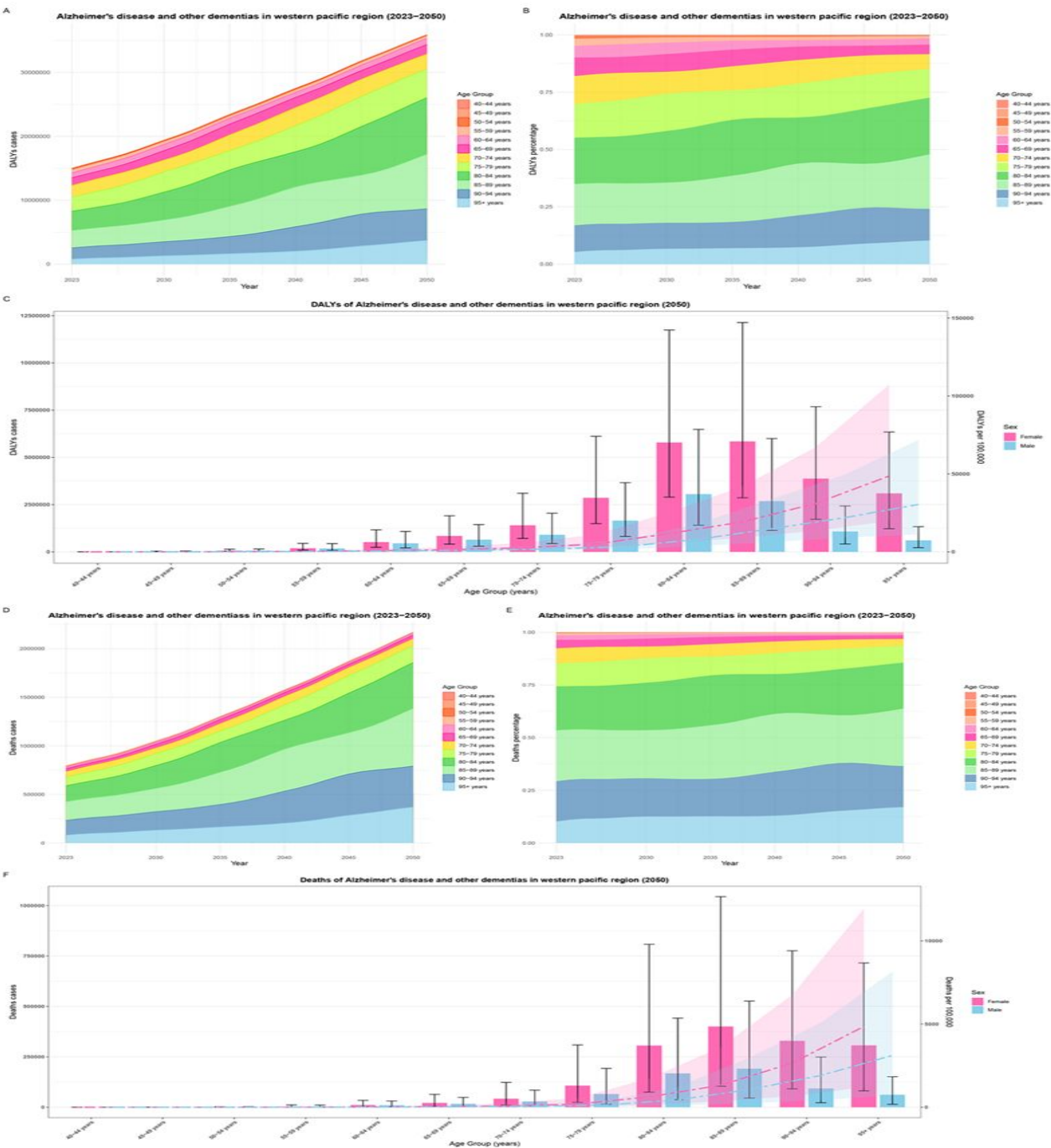


Figure 2. Estimated age-specific burden of Alzheimer's disease and other dementias in the Western Pacific region, 2023–2050. A) Absolute change in total DALYs (counts) by age group, both sexes; B) Percentage change in total DALYs (counts) by age group, both sexes; 2050 DALYs by age group and sex: bars = counts (left axis); dotted lines = DALY rates per 100,000 (right axis); 95% uncertainty intervals shown; C) Absolute change in deaths (counts) by age group, both sexes; D) Percentage change in deaths (counts) by age group, both sexes and F) 2050 deaths by age group and sex: bars = counts (left axis); dotted lines = death rates

per 100,000 (right axis); 95% uncertainty intervals shown. (DALY, disability-adjusted life year.)

Figure 3 ranks countries in the Western Pacific by projected rates in 2023 and 2050 and shows how positions change as dementia burden rises. For DALY rates, Japan remains the highest-burden country (from 2,086.4 to 2,945.9 per 100,000; +41.2%, rank 1→1), while the Republic of Korea climbs close behind in level (from 960.4 to 2,762.6; +187.6%, 2→2) and China moves into third (from 754.5 to 2,272.0; +201.2%, 5→3). Singapore shows a large relative increase (from 511.8 to 1,460.9; +185.5%, 8→4), whereas Australia and New Zealand rise more modestly (Australia 844.9→1,289.6; +52.6%, 3→6; New Zealand 829.1→1,351.0; +63.0%, 4→5). The steepest relative growth in DALY rate is observed in Brunei Darussalam (195.8→863.8; +341.1%, 19→11), followed by China (+201.2%) and Palau (320.1→946.7; +195.7%, 11→9). Several Pacific Island states change rank despite smaller absolute levels; for example, Samoa increases only +47.5% (rank 15→23), while Vanuatu and Papua New Guinea roughly double their DALY rates (Vanuatu +100.0%, 25→24; Papua New Guinea +115.1%, 27→26).

The death-rate panel shows a parallel pattern. Japan remains highest in 2050 (143.9→225.4; +56.6%, 1→1). Republic of Korea and China again show very rapid increases (54.9→196.2; +257.5%, 2→2 and 36.8→132.8; +260.5%, 5→3, respectively), and Singapore rises markedly (28.5→100.8; +254.0%, 8→4). Australia and New Zealand increase more slowly (+73.4%, 3→5; +84.5%, 4→6). The largest relative gains are seen in Brunei Darussalam (8.7→47.4; +444.7%, 18→11) and Palau (13.1→49.0; +273.0%, 13→9). Rank movements therefore identify countries where dementia burden is rising fastest relative to peers (e.g., Brunei Darussalam: +8 ranks in DALY rate, +7 ranks in death rate; Singapore: +4 in both), as well as settings whose relative standing declines despite absolute increases (e.g., Australia and New Zealand move down two places on both metrics).

Across the 28 Western Pacific countries/areas in our sample, DALY-rate ranks exhibit substantial churn from 2023 to 2050: 14 move up toward worse ranks, 11 move down toward better ranks, and 3 are unchanged (Japan = 1→1; Republic of Korea = 2→2; Kiribati = 23→23). The mean absolute rank change is 2.86 (median 2; range -8 to +9). The largest upward movers are Brunei Darussalam (20→11; +9 ranks), Singapore (9→4; +5), Lao People's Democratic Republic (22→18; +4), and Fiji/Palau/Philippines (each +3). The largest downward movers are Samoa (16→24; -8), Tuvalu (14→21; -7), Tonga (13→20;

–7), Tokelau (8→12; –4), and Australia (3→6; –3). Death-rate ranks display a similar pattern: 13 move up, 11 move down, and 4 are unchanged (Japan = 1→1; Republic of Korea = 2→2; Vanuatu = 25→25; Solomon Islands = 26→26), with a mean absolute rank change of 2.71 (median 2; range –8 to +8). The greatest upward movers for deaths are Brunei Darussalam (19→11; +8), Singapore (9→4; +5), Palau (14→9; +5), and Marshall Islands (27→23; +4); the largest downward movers are Tuvalu (13→21; –8), Samoa (15→22; –7), Tonga (11→16; –5), Tokelau (7→12; –5), and Nauru (24→28; –4). Ranks are computed from posterior mean rates; 95% uncertainty intervals are omitted for readability, and percentage changes can appear large in small populations because baselines are low and statistical uncertainty is wider.

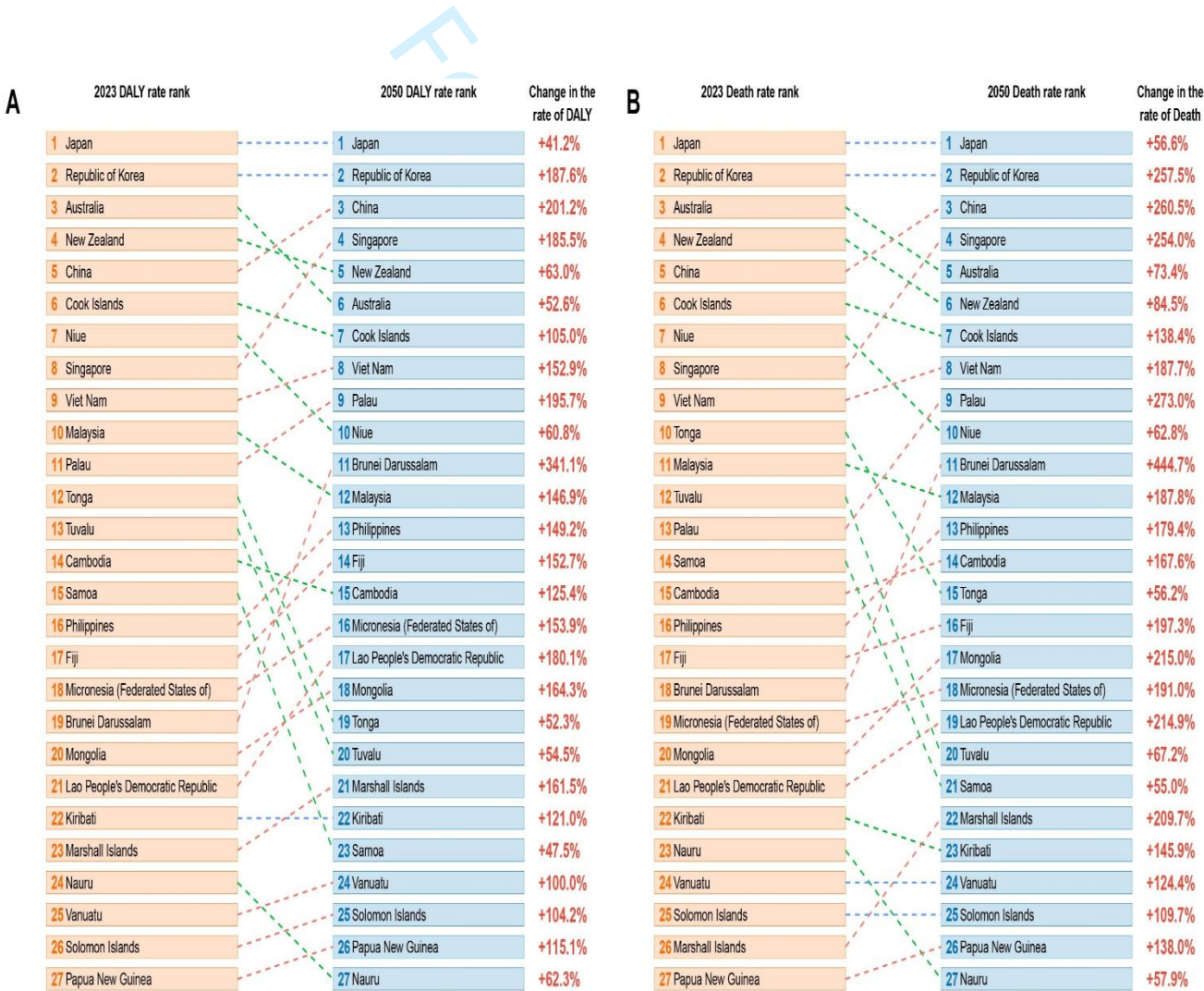


Figure 3. Estimated national-level changes in the DALY rate (A) and death rate (B) due to Alzheimer's disease and other dementias in the Western Pacific region, 2023–2050. Red dashed lines indicate rank increases, green dashed lines indicate rank decreases, and blue dashed lines indicate no change. DALYs = disability-adjusted life-yea

In Figure 4, the four choropleths display all-age rates per 100,000 for the Western Pacific; panels A–B show 2023 and panels C–D show 2050; class breaks and colour scales are held constant across years; values are posterior means (uncertainty not mapped).

(A) 2023 DALYs. The highest DALY burden clusters in the super-aged northeast: Japan sits at the top of the scale ($\approx 2,086$ per 100,000), while Republic of Korea (≈ 960) and China (≈ 755) form a contiguous mid-to-high band across East Asia. Australia and New Zealand map to intermediate classes (≈ 845 and ≈ 829 , respectively). Much of Melanesia, Micronesia, and Polynesia occupies the lower classes, indicating comparatively smaller all-age disability burden in 2023.

(B) 2023 deaths. The mortality pattern mirrors DALYs with an eastward apex: Japan again leads (≈ 144 per 100,000), followed by Republic of Korea (≈ 55) and China (≈ 37). Australia and New Zealand lie in mid classes (≈ 52 – 54), and most Pacific Island states remain in the lowest classes, yielding a north/east-to-south/west gradient in fatal burden.

(C) 2050 DALYs. By 2050 almost all of East and parts of Southeast Asia shift one to two colour classes higher, producing a contiguous high-burden corridor from China ($\approx 2,272$) through Korea ($\approx 2,763$) to Japan ($\approx 2,946$). Singapore moves decisively upward ($\approx 1,461$), while Australia and New Zealand rise more modestly and remain in lower classes relative to East Asia. Pacific Island states largely remain in the lowest classes despite sizeable relative growth, with Nauru, Papua New Guinea, Solomon Islands, and Vanuatu still at the bottom of the distribution.

(D) 2050 deaths. The death-rate map shows the same escalation: Japan remains highest (≈ 225), with Republic of Korea (≈ 196) and China (≈ 133) converging upward; Singapore crosses into a higher class (≈ 101). Australia and New Zealand increase but stay below the East Asian peak, and the Pacific Islands largely remain in lower classes. The identical scales across years make the eastward deepening of mortality visually evident and highlight a widening inter-country spread by mid-century.

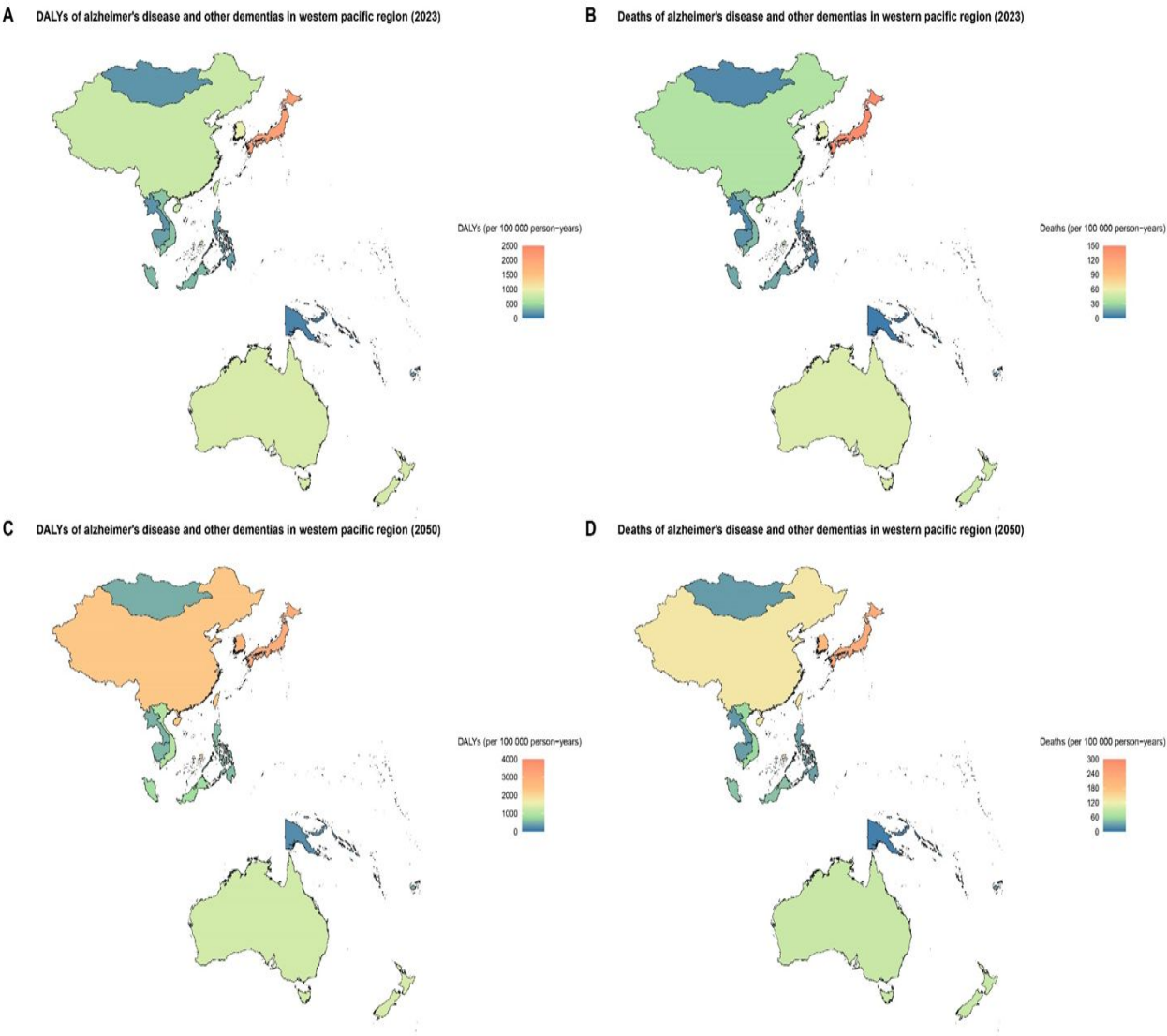


Figure 4. Estimated DALY rates (A, C) and death rates (B, D) per 100,000 population due to Alzheimer's disease and other dementias in the Western Pacific region in 2023 and 2050. DALYs = disability-adjusted life-years

Figure 5 panel A plots all-age DALY rates per 100,000, and panel B plots all-age death rates per 100,000, each under five alternative futures for the Western Pacific from 2023 to 2050.

Across both panels the message is straightforward—the burden rises steeply in every scenario. DALYs roughly double and deaths roughly triple by 2050. For most of the timeline the lines are almost on top of each other; they only start to separate in the early–mid 2040s. By 2050, the Combined package ends up highest, followed by Improved behavioural and metabolic risks, then Safer environment; the Improved childhood nutrition and vaccination path is virtually identical to the Reference path. The gaps between scenarios are small

compared with their uncertainty bands (on the order of a few tens of DALYs or deaths per 100,000), so the overall picture does not change with the scenario chosen.

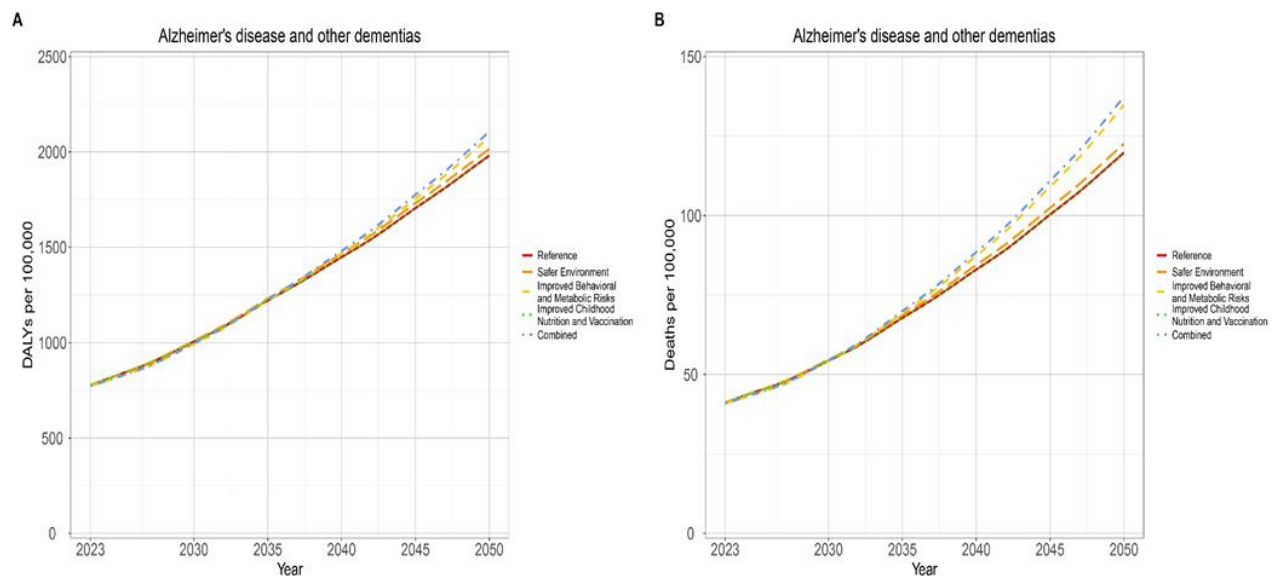


Figure 5. Estimated trends in DALY (A) and death (B) rates due to Alzheimer's disease and other dementias under different scenarios in the Western Pacific region, 2023–2050. DALYs = disability-adjusted life-year

The seemingly counter-intuitive result—that some “better” scenarios sit above the reference line—has a simple explanation: those packages help people live longer, which means more individuals reach their 80s and 90s where dementia is common and often fatal. Because these figures show all-age rates (not age-standardised rates), that longevity dividend raises the population-wide numbers even if age-specific risks improve modestly.

In practical terms, the figure says prevention and healthier environments still matter, but demography dominates the next 25 years. Health systems should plan for sustained growth in memory assessment, long-term and palliative care, and caregiver support, while continuing life-course risk-reduction whose benefits will accumulate beyond mid-century.

Discussion

The trajectories in Figure 1 are consistent with a region undergoing rapid demographic ageing and improved survival into the oldest ages, where dementia incidence, disability, and mortality are highest. International evidence indicates that the number of people living with dementia will continue to expand as populations age, with global counts projected to reach

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approximately 139 million by 2050 (Alzheimer’s Disease International, 2024). WHO likewise identifies dementia as a major and growing cause of dependency, disability, and death, with burden concentrated in older adults and disproportionately affecting women, largely because women live longer and accumulate more time at ages of highest risk (World Health Organization, 2023). Historical global analyses from the Global Burden of Disease program show higher dementia rates in women than men and document substantial increases in incidence, prevalence, mortality, and DALYs since 1990, with some evidence that the pace of increase has been faster in men in certain settings (Li et al., 2022; GBD 2019 Dementia Forecasting Collaborators, 2022). The Western Pacific projections here nevertheless show sustained female predominance and a widening absolute gap, which is plausible given the region’s super-aged populations (e.g., Japan, Republic of Korea) and its age–sex structures that channel more women into the oldest-old groups where dementia fatality and disability are greatest (Liu et al., 2025; Liu et al., 2024). The faster rise in death rates than DALY rates at the population level implies growing concentrations of very old individuals with dementia, fewer competing causes of death at earlier ages, and improved survival into age bands where case-fatality is higher. Because Figure 1 displays all-age rates rather than age-standardised rates, compositional ageing contributes to the upward trend; however, even accounting for this, the magnitude of increase signals substantial future demand for timely diagnosis, multidisciplinary dementia care, caregiver support, long-term care, and end-of-life services across the Western Pacific. These findings align with global calls to integrate dementia into universal health coverage and to pair late-life care planning with earlier-life risk reduction, including cardiovascular risk control, physical activity promotion, and education policies that build cognitive reserve (Alzheimer’s Disease International, 2024; World Health Organization, 2023; Li et al., 2022; Liu et al., 2025).

The age- and sex-specific patterns in Figure 2 indicate that the Western Pacific’s Alzheimer’s disease and other dementias (ADODs) burden will concentrate ever more heavily in the very old and in women. Absolute DALYs and deaths escalate sharply above age 70, with the largest gains at 80–94 years and ≥ 95 years, consistent with global evidence that increases in dementia prevalence, mortality and DALYs over recent decades are driven primarily by population ageing and longer life expectancy rather than abrupt deterioration in age-specific risks (Li et al., 2022). The same literature shows a persistent female predominance—women comprise the larger share of people living with dementia worldwide—largely because they

survive to the oldest ages where incidence is highest (Li et al., 2022; World Health Organization [WHO], 2023).

Panels A–B show that nearly all future growth in DALYs arises from cohorts entering ≥ 75 years, with declines at ages 40–59 reflecting demographic composition rather than falling age-specific risk. Panel C demonstrates a marked compositional shift by 2050: ages 80–94 and ≥ 95 together account for roughly three-quarters of DALYs, while ≤ 69 contribute a small minority. Panels D–E show the same ageward re-allocation for deaths, with the steepest percentage growth in the oldest-old. Panel F then highlights two implications for service planning: (i) dementia mortality becomes heavily concentrated in ≥ 80 -year-olds and (ii) age-specific DALY and death rates are consistently higher in women, mirroring global summaries that also note women provide most informal dementia care—about 70% of care hours—intensifying the gendered social and economic impact (WHO, 2021, 2023).

Overall, age remains the dominant risk factor for dementia, but modifiable risks shape the level and timing of burden. WHO guidance and burden studies identify hypertension, diabetes, obesity, smoking, harmful alcohol use, physical inactivity, depression, social isolation and related factors as contributors to dementia risk, and they recommend life-course prevention to delay onset and compress disability (WHO, 2019, 2023; Li et al., 2022). Within GBD analyses, high body-mass index, high fasting plasma glucose and smoking are prominent risk contributors to ADODs, helping to explain sex-specific patterns and the steeper growth observed in very old ages as cardiometabolic exposures accumulate across the life course (Li et al., 2022). Together, these findings imply that, while prevention remains necessary, demographic momentum will dominate the next quarter-century; health systems should therefore expand dementia-capable primary and geriatric care, long-term and palliative services, and implement gender-responsive policies that ensure equitable access to diagnosis and care and targeted support for female caregivers and older women living alone (WHO, 2021, 2023).

The national rank shifts in Figure 3 reflect how population age structure, survival, and system preparedness interact across the Western Pacific. Super-aged societies (Japan; Republic of Korea) remain at the top because a large share of their populations has already reached very old ages, where dementia incidence and case fatality are highest (Gong et al., 2024). Rapid ascents for China and Singapore are consistent with large middle-aged cohorts ageing into high-risk bands, improving survival from cardiovascular and infectious causes, and expanded

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detection/attribution as health systems mature—dynamics that global syntheses anticipate as primary drivers of rising all-age burden even when age-specific risks change only gradually (ADI, 2024; Li et al., 2022). Outliers such as Brunei Darussalam (+large percentage gains from a low baseline) and Palau illustrate how small populations can register sharp relative increases as longevity improves, while Australia and New Zealand drift down in *rank* despite absolute growth, reflecting faster increases among their neighbours.

Policy readiness is uneven. Only **six** Western Pacific economies have standalone national dementia plans (Australia, Japan, New Zealand, Republic of Korea, Taiwan, Singapore), and five have integrated plans; most countries have no formal strategy (Timmins et al., 2024). This gap helps explain why upward movers often coincide with under-prepared systems. Planning should therefore proceed on two tracks: (i) sustain and adapt capacity in long-standing high-burden settings (e.g., Japan) for an increasingly complex, very old case-mix, and (ii) accelerate scale-up of diagnosis, community-based long-term care, caregiver support, and palliative services in rapidly rising countries (e.g., China, Singapore, Brunei).

Figure 4 maps all-age DALY and death rates per 100,000 for 2023 (A–B) and 2050 (C–D) using identical class breaks and colour scales, with values shown as posterior means. In 2023, the highest DALY and death burdens cluster in the super-aged northeast—Japan at the apex, with Republic of Korea and China forming a contiguous mid-to-high corridor—while Australia and New Zealand sit in intermediate classes and most Pacific Island states occupy lower classes (Gong et al., 2024). By 2050, the gradient steepens and shifts eastward: East Asia moves one to two classes higher, producing a high-burden corridor from China through Korea to Japan; Singapore rises markedly, whereas Australia and New Zealand increase more modestly and the Pacific Islands largely remain in lower classes despite sizeable relative growth (Gong et al., 2024). This cartographic deepening is fully consistent with demographic transition in the region, where the population aged ≥ 65 is large and projected to expand sharply, concentrating dementia in the oldest-old (Li et al., 2022). It is also compatible with the region’s risk profile: many Western Pacific countries face above-global-average exposure to diabetes, obesity, alcohol use, and physical inactivity, alongside hypertension and smoking—factors linked to later-life dementia (Gong et al., 2024; Li et al., 2022; World Health Organization [WHO], 2019).

Because these are all-age rates, much of the 2023→2050 intensification reflects compositional ageing—more people surviving into ages where dementia incidence and case-

fatality are highest—rather than abrupt deterioration in age-specific risk (Li et al., 2022). We therefore recommend presenting age-standardised companion maps in the Supplement to separate risk change from population structure; nevertheless, the 2050 cross-section still indicates substantially higher burden in East and parts of Southeast Asia, and persistent, though lower, burden in younger Pacific Island populations (Gong et al., 2024). Policy implications follow directly. Super-aged, high-burden systems (Japan; Republic of Korea) should sustain and adapt capacity in memory assessment, dementia-capable primary and geriatric care, long-term and palliative services, and caregiver support, with attention to the disproportionate burden borne by women (WHO, 2021, 2023). Rapidly rising settings (China, Singapore and selected middle-income countries) should accelerate scale-up of diagnostic and long-term-care infrastructure while investing across the life course in risk-reduction (cardiometabolic control, tobacco and harmful-alcohol reduction, physical activity, hearing care, and education that builds cognitive reserve) to delay onset and compress disability (WHO, 2019; WHO, 2023). Given that only a minority of Western Pacific countries have standalone or integrated national dementia plans, closing planning and implementation gaps—particularly in Pacific and lower-resource contexts—remains urgent (Timmins et al., 2024).

The age-specific forecasts in Figure 5 highlight that Alzheimer's disease and other dementias (ADODs) will impose a steeply rising burden in the Western Pacific through 2050 under every scenario. All-age DALY rates roughly double and death rates nearly triple across the period, with scenario lines staying virtually superimposed until the early–mid 2040s.

Panels A and B demonstrate that Alzheimer's disease and other dementias (ADODs) will impose a steep, persistent rise in population-level burden across the Western Pacific under all five scenarios to 2050: all-age DALY rates approximately double and death rates nearly triple, with scenario curves largely overlapping until the early–mid 2040s and only modestly diverging thereafter. By 2050, the combined package sits highest, followed by improved behavioural/metabolic risks and safer environments, while the improved childhood nutrition/vaccination path is effectively indistinguishable from the reference—differences that are small relative to wide uncertainty intervals and therefore do not alter the qualitative trajectory (Gong et al., 2024; Li et al., 2022).

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The apparently counter-intuitive ordering—where “better” scenarios yield slightly higher all-age rates—reflects composition: improvements in survival and early-life conditions expand the number of people living into their 80s and 90s, ages at which dementia incidence and case-fatality are highest. Because Figure 5 reports all-age (not age-standardised) rates, this longevity dividend elevates population-wide levels even if age-specific risks improve modestly. This interpretation is consistent with regional demographics—over 245 million people aged ≥ 65 already live in the Western Pacific and that number is expected to double by 2050—and with global evidence attributing much of the rise in dementia prevalence, mortality, and DALYs to population ageing and longer life expectancy rather than abrupt shifts in age-specific risk (Gong et al., 2024; Li et al., 2022).

These demographic pressures interact with a high baseline of modifiable risks in the region. Many Western Pacific countries face above-global-average prevalence of alcohol use, diabetes, obesity, and physical inactivity, while hypertension, high fasting glucose, smoking, and related metabolic factors remain established contributors to late-life dementia (Gong et al., 2024; Li et al., 2022; World Health Organization [WHO], 2019). Women bear a disproportionate burden of dementia and provide most informal care hours, intensifying the gendered impact of rising late-life need (WHO, 2021, 2023). Accordingly, prevention and healthier environments remain essential—to delay onset and compress disability—even though they cannot, on their own, reverse the demographic tide before mid-century (WHO, 2019; WHO, 2023).

For policy, the message is two-handed. First, scale capacity for memory assessment, dementia-capable primary and geriatric care, community and residential long-term care, palliative and end-of-life services, and structured caregiver supports—explicitly incorporating gender-responsive approaches (WHO, 2021). Second, expand life-course risk-reduction (tobacco control; cardiometabolic risk management; physical activity; hearing care; social connection; education that builds cognitive reserve) in line with WHO guidance, recognising that benefits accumulate beyond 2050 (WHO, 2019; WHO, 2023). To sharpen interpretation, present age-standardised scenario curves alongside the all-age series and, where feasible, decompose growth into age-structure versus age-specific rate components; these additions clarify how much of the increase is demography versus risk, and they can guide priority-setting (Li et al., 2022). Finally, because only a minority of Western Pacific countries currently have standalone or integrated national dementia plans, regional

collaboration and targeted investment in dementia literacy, diagnosis, and care remain urgent (Timmins et al., 2024).

Conclusion

Across the Western Pacific, Alzheimer's disease and other dementias (ADODs) will accelerate from a substantial problem to a defining challenge of healthy ageing over the next quarter-century. All-age DALY rates approximately double and death rates nearly triple by 2050, with the steepest absolute increases concentrated in the oldest-old and a persistent female predominance. This pattern is fundamentally demographic: longer lives and large cohorts moving into ages where dementia incidence and case-fatality are highest magnify population-level burden even when age-specific risks improve modestly. National trajectories diverge: Japan and the Republic of Korea remain high-burden systems, while China and Singapore record the largest absolute gains; Pacific Island states generally retain lower rates but still rise meaningfully. Scenario analyses reinforce the same conclusion: improving environments, cardiometabolic profiles, and early-life conditions slows—but does not reverse—the upward arc because survival gains enlarge the number of people living into high-risk ages. Taken together, these findings imply sustained growth in demand for dementia prevention, timely diagnosis, long-term and palliative care, and structured caregiver supports, with particular attention to women and to countries where rates and ranks rise fastest.

Limitations

This study uses scenario-based forecasts built on GBD-2021 aggregated estimates. As with any secondary analysis, results depend on the completeness and comparability of the underlying data. Dementia is underdiagnosed in many settings and diagnostic coverage, case ascertainment, and coding practices vary across countries and over time; stigma and cause-of-death misclassification can further bias levels and trends. We report uncertainty intervals, but these reflect sampling and model error rather than all potential sources of systematic bias. National estimates may mask substantial subnational heterogeneity, and projections for small island states—where data are sparse—are inherently less precise than for large populations.

The forecasts themselves rest on structural assumptions about future covariates and scenario paths. We model exogenous improvements in environmental, behavioural, metabolic and

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early-life risks, but do not endogenise policy uptake, programme fidelity, or non-linear interactions among risks. Our projections also cannot anticipate disruptive changes—such as widespread biomarker-anchored diagnostics, disease-modifying therapies, shifts in competing mortality, migration, pandemics, or climate-related shocks—that could alter incidence, survival, or attribution. Because primary figures emphasise all-age rates, part of the observed rise reflects compositional ageing rather than changes in age-specific risk; age-standardised summaries mitigate but do not eliminate this limitation. The model quantifies statistical uncertainty via draws, yet structural uncertainty about model form and covariate selection is harder to capture.

Finally, our study focuses on the composite cause “Alzheimer’s disease and other dementias” and does not project subtype-specific trajectories that might carry distinct prevention and service implications. We did not quantify economic costs, caregiving intensity, or access to and quality of dementia care, nor did we stratify by education, rurality, or socioeconomic status; these omissions limit conclusions about equity and financing needs. Collectively, these constraints argue for cautious interpretation of point estimates and for using the projections as directional planning inputs rather than precise forecasts.

Recommendations

Policy must proceed on two tracks that recognise both prevention’s value and demography’s momentum. First, governments should adopt or update fully costed national dementia plans that embed prevention, timely diagnosis, treatment, long-term and palliative care, and caregiver support within universal health coverage. Plans should be calibrated to country-specific trajectories—rapid capacity expansion where growth is fastest (e.g., China, Singapore) and sustained adaptation in systems already at high burden (Japan, Republic of Korea)—and should include gender-responsive provisions given women’s higher prevalence and caregiving roles. Second, health systems should expand life-course risk-reduction at scale—tobacco control; hypertension, diabetes and lipid management; promotion of physical activity and healthy weight; hearing care; social connection; and education that builds cognitive reserve—recognising these measures delay onset and compress disability even if they cannot fully offset ageing by 2050.

Service delivery must be re-tooled for sustained demand. Primary care and geriatrics should be equipped for cognitive screening and longitudinal management; multidisciplinary

pathways spanning neurology, psychiatry, rehabilitation and social care should be standard; and dementia-capable home- and community-based services and residential long-term care should be scaled with clear staffing, quality and equity benchmarks. Financing reforms that cover long-term care and provide structured caregiver supports—training, respite, psychosocial services and protection from catastrophic spending—are essential to maintain family care while reducing uncompensated burden. To steer policy, countries should strengthen surveillance and registry systems; report both age-standardised and all-age indicators with uncertainty; disaggregate by sex and subnational region; and commission comparative- and cost-effectiveness studies of care models, including technology-enabled monitoring and assistive solutions. Finally, regional collaboration in the Western Pacific—shared indicators, pooled implementation research, and policy transfer on long-term-care financing and workforce development—can compress learning cycles and improve preparedness across diverse settings.

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